COMP34812 Natural Language Understanding Coursework (AY 2025-26)

I. Introduction

For the COMP34812 coursework, your group (with two members) will participate in what we call a *shared task.* In the natural language processing (NLP) community, shared tasks are often organised to assess state-of-the-art performance on a particular task. There is a shared task for almost any NLP problem, but some of the more recently organised ones include the SemEval 2023 task on Multilingual Complex Named Entity Recognition[1], the BioNLP 2024 task on Lay Summarization of Biomedical Research Articles[2] and the Multilingual Lexical Simplification task[3], to mention a few examples.

A dataset is (almost always) provided to participants in a shared task, often consisting of training and development (also known as "validation") subsets. Tasks can be either *open* or *closed*; the former means that participants are allowed to exploit any resources/datasets outside of what was officially provided by the task organisers, while the latter constrains the participants to using only the dataset provided as part of the shared task.

A shared task's official evaluation (test) data is often not made available to participants until much later, when enough time has passed such that the participants are likely to have completed the development of their solution(s). Once the test data is released, participants are typically given just a short time to run their solution(s) on the test data in order to generate outputs, i.e., predictions, according to an agreed format. The participants then submit the predictions produced by their solution(s), which are evaluated against the hidden, gold standard annotations (labels) in the test set, using a number of metrics. The performance of the submitted solution(s) is finally reported in terms of those metrics.

Sometimes, a shared task consists of multiple *tracks*, each of which is focussed on a specific problem, domain or subtask. In the NLU coursework, we are organising a shared task on **pairwise sequence classification** with three different tracks. Your group will choose the one track that they wish to participate in.

II. Intended Learning Outcomes

- To build and implement two different solutions to a pairwise sequence classification problem
- To describe and present your solution in the form of a flash presentation
- To act as a responsible project partner, collaborate on project development/writing up, and contribute to planning/organisation

III. The Shared Task and its Tracks

The NLU shared task is focussed on pairwise sequence classification, the task of categorising a pair of sequences according to a predefined set of classes that capture the relationship between the two sequences. The three tracks that we are organising are:

A. Natural Language Inference (NLI)

[1] https://multiconer.github.io/
[2] https://biolaysumm.org/
[3] https://sites.google.com/view/mlsp-sharedtask-2024/home

Given a *premise* and a *hypothesis*, determine if the hypothesis is true based on the premise. You will be given more than 24K premise-hypothesis pairs as training data, and more than 6K pairs as validation data.

B. Evidence Detection (ED)

Given a *claim* and a piece of *evidence*, determine if the evidence is relevant to the claim. You will be provided with more than 21K claim-evidence pairs as training data, and almost 6K pairs as validation data.

C. Authorship Verification (AV)

Given two sequences, determine if both sequences were written by the same author. You will be provided with more than 27K sequence pairs as training data, and 6K sequence pairs as validation data.

It is worth noting that we are running the shared task in **closed** mode, hence, it is not permissible to use any existing datasets outside of those that we are providing, in developing your solutions.

IV. Instructions

This section provides an overview of what needs to be done as part of the coursework. Details pertaining to deadlines, output formats and submission sites are provided in succeeding sections.

1. Identify one or two project partners. As a group, consider the three available tracks and decide which track you wish to participate in. You might find it useful to have a look at a small trial dataset that contains examples of sequence pairs and their corresponding labels.

Once you have made a choice, register your group and your chosen track by completing an online form; this form needs to be completed by just one representative per group. Note that the online form is set up in such a way that you can edit your responses until the deadline (see Section VII); however, after this deadline, you will not be able to switch to another track.

2. Develop **two** different solutions to the problem that you have chosen, where any two of the following categories of approaches should be represented:

A. Unsupervised or traditional machine learning-based approaches
B. Deep learning-based approaches that do not employ transformer architectures
C. Deep learning-based approaches underpinned by transformer architectures

IMPORTANT NOTES:

- For each of the three tracks, we (the marking team) have developed baseline methods, one from each of the three categories of approaches above. You will likely obtain a passing mark even if your solution is unable to outperform the baseline performance, but higher marks will be given to solutions that obtain improved performance (in comparison with the baseline methods).
- You are free to choose any method/model that you wish to develop, and will likely obtain a passing mark even with just standard approaches (e.g., fine-tuning a BERT-based model). However, as can be seen in the Marking Rubric (Section IX), higher marks will be given to more creative/inspired solutions.
- Your two solutions must come from **two different approach categories** (A/B/C).Simple variations within the same category (e.g., swapping BERT variants or minor hyperparameter changes) do **not** count as a second solution.

3. Once the test data is released, run each of your developed solutions on it and produce predictions. Note that the test data will follow the same format as the files in the trial dataset, except that the **label** column will not be provided.

Submit the **predictions** in the required format by the coursework deadline. Your predictions will be evaluated against the (hidden) gold standard labels in the test set, using standard metrics such as F1-score. See Deliverable 1 in Section V.

4. Prepare the code for your solutions by the coursework deadline. Specifically, there should be code that can run each of your solutions in "inference mode", henceforth referred to as **demo code**: given an input file containing test data, the code should run the solution and generate the required predictions file. Overall, the code should be well-documented so as to help the marking team in identifying the code that corresponds to the training/building of each of your chosen approaches. See Deliverable 2 in Section V.

IMPORTANT NOTE: The demo code should be in the form of a **Python notebook** that contains lines for installing any required packages, to allow the marking team to run it on their own.

5. Create a **model card** for each of your solutions. A model card[4] is a document, typically a markdown file, that concisely describes a model. When sharing models with people outside of your own group, it is good practice to provide some key information that will help make your work reusable. To help you in creating your model cards, a model card template and a Python notebook for generating your customised model cards have been provided. Your generated model cards should be included in your code submission. See Deliverable 3 in Section V.

6. Prepare a **poster** that describes your solutions and how you developed them. This poster will be the basis of your flash presentation to the marking team during Workshop 4 (see Section VII), that both you and your partner are required to attend in person. Additionally, during the flash presentation you will showcase each of your solutions in real time using your demo code (see Item 4 above).

- **Poster format:** single-page PDF (A1 landscape) *or* one-slide PDF exported from PowerPoint (16:9).
- **Flash presentation:** 3-4 minutes presentation of poster + 3-4 minutes live demo+ 3-4 minutes Q&A.
- Poster must include at minimum: task/track, dataset summary, both methods, dev results table, brief error analysis, and limitations/ethical considerations.

V. Deliverables and Submission Sites

Deliverable 1. The predictions produced by each of your two solutions on the test data. These will be submitted via Canvas. Please name your CSV files according to the following convention: *Group_n_category.csv*, where *n* is your group number (which you will find in Canvas) and *category* is one of *A*, *B* or *C* (in uppercase), depending on the category that the solution (that produced the predictions) belongs to. See Section IV, Item 2 above for the list of categories.

Deliverable 2. Your code for developing/training models and evaluating them (e.g., on the development set), as well as demo code, i.e., runnable code and models for generating predictions for a given input data file, in the form of a Python notebook. It is expected that your submission is organised or structured in such a way that:

[4] https://www.kaggle.com/code/var0101/model-cards

- any code that was written to train models should be separated from the demo code; make sure that your models have been saved and can be readily loaded by the demo code;
- any code for evaluation (on the development set) is included; this should be separated from your code for training models
- someone else outside of your team can follow the code; in other words, provide some documentation, i.e., in-line comments and a README explaining the code structure and how it is run.

Importantly, your README should provide: (1) attribution to any data sources you used or code bases you reused, and (2) links to any models that you yourself trained and stored on the cloud (i.e., OneDrive).

IMPORTANT NOTES:

- Failure to explicitly acknowledge that you reused (parts of) someone else's code will be treated as academic malpractice.
- If any of your resources (e.g., models) are more than 10MB, please do NOT include them in your Canvas submission and instead store them on the cloud.

Deliverable 3. A model card for each of your solutions, in markdown format.

IMPORTANT NOTE: All deliverables (Deliverables 1, 2 and 3) should be compressed into **one zip file**, which should then be uploaded to the relevant Canvas submission site.

VI. Use of Generative AI Tools

The University of Manchester has recently published Guidelines for Staff and Students Using or Developing AI. Students are reminded that within the context of this coursework, the following section of the university's guidelines applies to each of the coursework deliverables: your code, model cards and poster.

"There is an expectation that work submitted by a student for assessment should contain their own original work and provide an honest representation of their understanding of the subject. ***Students should declare any use of generative AI in preparing work for assessment, and explain its role.*** *Submitting work created by Generative AI as their own, or to misrepresent their understanding of the subject, is plagiarism and will be dealt with in accordance with the University's Academic Malpractice Procedure."*

If your team has used generative AI tools in preparing/developing your coursework, please declare this in the following way: in your README, add a section called "Use of Generative AI Tools" and describe which generative AI tools were used and for what purposes.

VII. Release of Data, Timetabled Activities and Deadlines

Highlighted in red font below are critical deadlines/activities that every group should definitely pay attention to.

Event	Description	Site	Relevant Date
Release of coursework specifications and data	Trial data for all tracks published	Canvas (look for "Trial Data for Shared Task")	26 Feb 18:00
Workshop 1	Q&A regarding coursework	In-person	05 March 09:00-10:00

	specifications; groups should be finalised		
Deadline: Registration of groups	Groups register their members and their chosen track	Microsoft Form	05 March 18:00
Workshop 2	Overview of the Codabench platform; Q&A	In-person	12 March 09:00-10:00
Workshop 3	Advice on preparing the deliverables, including model cards, for submission	In-person	19 March 09:00-10:00
Release of test data	Test data for all tracks published	Canvas	24 March 18:00
Deadline: Submission of all deliverables	Groups upload their system predictions, code, and model cards	Canvas for all deliverables: system predictions, code, and model cards	27 March 18:00
Presentation: Present your poster and demo code	Flash presentation + running demo code	In-person	29 Apr; 12:00-13:00 30 Apr; 9:00-10:00

VIII. Teamwork

Each team member will be given the same mark for the coursework. One of the Intended Learning Outcomes of this coursework is focussed on acting as a responsible team member (see Section I), hence it is each team member's duty to ensure that tasks are delegated fairly and that there are equal contributions.

IX. Marking Rubric

The coursework is worth 40 marks overall. The table below outlines the criteria comprising the marking rubric and the distribution of marks across these criteria.

System Predictions		
Competitive performance (Solution 1)	The solution obtains performance improvement that is statistically significant, in comparison with a baseline method for the same type of approach.	3
Competitive performance (Solution 2)	The solution obtains performance improvement that is statistically significant, in comparison with a baseline method for the same type of approach.	3

Implementation of Solutions		
Organisation and documentation	The code is well-documented and structured in a way that even developers outside of the team can follow it.	3
Completeness and Reproducibility	All necessary resources, including trained models and demo code, were provided for both solutions, allowing for reproducibility.	3
Soundness (Solution 1)	The design of the solution is technically sound; there were no obvious technical details that were ignored/misunderstood.	3
Soundness (Solution 2)	The design of the solution is technically sound; there were no obvious technical details that were ignored/misunderstood.	3
Creativity (Solution 1)	The design of the solution draws from literature on state-of-the-art approaches; the group was adventurous in that they went beyond the typical, standard approaches to the problem.	3
Creativity (Solution 2)	The design of the solution draws from literature on state-of-the-art approaches; the group was adventurous in that they went beyond the typical, standard approaches to the problem.	3
Evaluation	There is evidence that the group put effort into evaluating their solutions, outside of the supporting benchmarking tool (Codabench).	3
Model Cards		
Formatting	Both model cards were prepared in the correct format	3
Informativeness (Solution 1)	The model card sufficiently describes the model, allowing any potential users to reuse/build upon the model.	3
Informativeness (Solution 2)	The model card sufficiently describes the model, allowing any potential users to reuse/build upon the model.	3
Accurate representation	The model cards accurately represent the implemented solutions.	4